McKean-Vlasov models

Theory (1960-1990: McKean, Hitsuda, Méléard, …)

$$\left\{ \mathrm{d}X_t^i = b(X_t^i, \mu_t^N)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}W_t^i \right.$$

- Mean-field scaling: $a = O(1/N)$

- $\mu_t^N \to \mathscr{L}(X_t) \hat{=} \mu_t$ in law as $N \to \infty$

- μ_t solves McKean-Vlasov PDE $\partial_t \mu_t = \nabla_x \cdot (\sigma \nabla_x \mu_t - b(x, \mu_t)\mu_t)$

- $\sqrt{N}(\mu_t^N - \mu_t)$ converges to a Gaussian measure-valued process

- On bounded domains $t \in [0,T]$ and chaotic initial data $\mu_0^N = \mu_0^{\otimes N}$

- Coefficients with bounded derivatives

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Recent theory (2000s-now)

- (Non-)uniform-in-time convergence
- Singular coefficients
- Hypo-elliptic diffusions
- Initial correlations
- Non-exchangeable interactions